KIRLOSKAR 3GK550ETA 4SR1 · GK3.8703 · 25 KVA

Replacement Engine Control Unit

Complete documentation: the machine, the circuits, the specification, and the research all of it rests on.

This replaces the **Engine Control Unit** block on page 1 of the Kirloskar troubleshooting wiring diagram, manual 08-GP3-60-001. Every pin, signal and colour in these pages was read out of that document or measured in simulation; where something is inferred rather than confirmed, it says so on the line.

The engine has not been inspected yet. Of twelve items on the unknowns register, two are answered by inference and none is confirmed — every confirmation needs the machine. The CPCB IV+ certification position in Memo 06 is a decision for the machine's owners, not an engineering conclusion. **Read that memo before fitting anything.**

CONTENTS

1. **Build Status**
   What is done, what is not, and what each gap is waiting on
2. **Handoff**
   Setting up on another machine, what is verified, what is open, the traps
3. **Field Handbook**
   The machine, its interface, the circuits in narrative, and how to run it
4. **Sensor Reference**
   Every sensor: what it measures, how we read it, what we do with it
5. **Circuit Reference**
   Every circuit block: schematic, reasoning, simulated result, limits
6. **Reconciled Specification**
   Binding source of truth -- pin map, power, deviations, unknowns
7. **MCU Pin Assignment**
   S32K148 to the 94-way connector -- 37 signals, with package pins and reset state
8. **BOM Requirements**
   What the schematic has to buy, and the simulation block that says so
9. **Design Decisions**
   Every fork put to the owner: the question, the options, the choice, the reason
10. **Firmware Guide**
    How the engine-control software works: timing, safety, CAN, logs, knobs, tests, port

11. **PCB Layout Guide**
    For the person routing the board: classes, the ground join, each block's critical loops
12. **Memo 01 -- Engine Identity**
    Which engine this actually is, established from five hardware constraints
13. **Memo 02 -- KG640C Controller**
    What the existing genset controller does, and what it needs from us
14. **Memo 03 -- ECU Connector**
    The 94-pin connector: candidates, sourcing, and what stays unconfirmed
15. **Memo 04 -- Injector Drive**
    Bank sharing proved safe, and the drive envelope it has to hit
16. **Memo 05 -- MCU Selection**
    Why the S32K148, and the timing argument that actually binds
17. **Memo 06 -- Standards & Law**
    ISO, CISPR and the CPCB IV+ certification position
18. **Memo 07 -- BOM & Sourcing**
    Costed bill of materials, India sourcing, single-source flags
19. **Memo 08 -- Engine Inspection Brief**
    The measurements to take at the machine, grouped by machine state
20. **Memo 09 -- Resistive Sensor Front-Ends**
    How DSE and SEDEMAC build temperature inputs, and what transfers
21. **Memo 10 -- Injector Turn-Off**
    What production drivers do with the coil energy, and the circuit to build
22. **Memo 11 -- Supervisor & Fail-Safe Outputs**
    Watchdog, brownout, and what the outputs do while the MCU is in reset
23. **Memo 12 -- Driver Architecture**
    Integrated driver ICs versus discrete FETs, and why the rail decides it
24. **Phase 1 Research Plan**
    The plan the eight memos were written against